Operational semantics for Prolog with Cut in Rocq and its application to determinacy analysis

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Abstract

We mechanize two operational semantics for Prolog with cut and prove their equivalence in the Rocq prover.

The first semantics is the one introduced by Debray and Mishra [17] for studying the optimizations employed by Prolog abstract machines [36, 45]. This semantics is based on a stack of alternatives, where the cut operator is represented by a pointer to a shorter stack.

The second semantics is novel and based on And-Or trees that make the structure of the alternative computations explicit and preserve it throughout execution. On this structure the cut operator is characterized explicitly, as a function that prunes selected branches of the tree. Cut-tracking, in the style of [34, 37], then lets us state introduction and elimination rules for disjunction under cut, which, as expected, differ from the classical ones.

As a case study, we use the tree-based semantics to prove the soundness of the determinacy analysis introduced in [1]. By exploiting the equivalence between the two semantics, we relate this result to the stack-based semantics used in efficient implementations of logic programming such as [19].

Keywords: Semantics, Prolog, Elpi, Determinacy Analysis, Rocq, Coq

1 Introduction

Backtracking is a compelling feature of logic programming, but it is also a double-edged sword. On the one hand, it makes it possible to write concise code that searches for a solution or enumerates all solutions. On the other hand, backtracking is often a source of poor performance and can make logic programs harder to debug. To mitigate these problems, the logic programming community has studied various forms of determinacy analysis [1, 18, 23, 24, 27, 28, 40] that identify logic programs that are deterministic, i.e., that compute functions rather than relations: they commit to their first solution and leave no *choice points* (places where backtracking could resume). A choice point corresponds to a suspended *alternative* computation; in the following, we use *choice point* and *alternative* interchangeably. The control operator *cut* allows the programmer to discard choice points and is therefore pervasively used to write deterministic code, but it is also well known to make the semantics of the programming language depart from a purely logical reading, thereby making reasoning about code harder.

Intuitively, nondeterminism arises when a computation can be completed using two different rules. This situation is typically ruled out statically by performing two complementary checks. The first one, called *mutual exclusion*, checks that at most one rule can apply to a given query. This condition is satisfied if either (i) no two rules have overlapping heads, so that no query can unify with both, or (ii) the analysis accounts for the presence of cut in rule premises, so that once the cut is executed all remaining rules are discarded. The second check, called *rule-body checking*, verifies that each rule either (i) calls only functions, ensuring that no choice point remains after the premises of the chosen rule have been executed, or (ii) executes a cut after calling relations, in which case any choice points left by the call are discarded.

Most papers in the literature only provide paper proofs for the soundness of their analysis, with the exception of Kriener and King [27]. They mechanize in the Rocq prover (formerly the Coq proof assistant) a denotational semantics for Prolog with cut and use it to prove the soundness of an abstract-interpretation-based determinacy inference algorithm. In addition to their Rocq code being unavailable, this mechanization has two further shortcomings. First, the analysis does not scale to dynamic predicates (see [1, Sec. 8]), so it cannot be applied to λprolog [32, 33] which is our long term goal. Second, there is no mechanized link between their denotational semantics and the operational semantics of Debray and Mishra [17], which underlie standard Prolog abstract machines [4, 36, 45].

Contributions. In this paper we mechanize in Rocq two operational semantics for Prolog with cut. The first is the stack-based semantics given by Debray and Mishra [17]. This semantics is closely related to actual Prolog implementations, but it makes reasoning about the scope of cut difficult: alternatives are stored in a stack, in chronological order, and the cut operator is represented as a pointer to a shorter stack. The relation between the rules being cut and the affected stack frames is therefore not immediately visible.

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For reasoning about cut we design and mechanize a second semantics, based on And-Or trees [25], that makes the history of the computation explicit: for a given goal, each applicable rule yields an Or-branch, and each Or-branch carries one And-branch per premise of the rule. Unlike the computation trees of Börger and Rosenzweig [8, 9] and the SLD-trees used by Apt [3], which branch only on clause alternatives and keep the remaining goals as a flat sequence in each node, our trees also branch on the premises of a rule. On this structure the cut operator is characterized explicitly, as a function that prunes selected branches of the tree, and its scope is read off the position of the cut node, so no cutpoint pointer is needed. Following Mogensen [34] and Rozplokhas et al. [37], each execution additionally reports whether a cut was executed; this information lets us formulate introduction and elimination rules for disjunction in the presence of cut which, as expected, differ from those of classical logic: for instance, $q_1 \vee q_2$ is not equivalent to $q_2 \vee q_1$, since q_2 may be cut away by q_1.

Finally, we use the tree-based semantics to prove the soundness of the determinacy analysis from [1]. In Rocq, we prove that any terminating execution of a predicate identified as deterministic introduces no additional choice points. Combined with the equivalence between the two semantics, this result implies that any call to a deterministic predicate that terminates leaves no choice points on the stack used by actual Prolog implementations, including the Elpi [19] λprolog interpreter, for which the static analysis of [1] was originally developed. Elpi serves as an extension language for the Rocq prover and has become an important component of the Rocq ecosystem [7, 11–13, 20–22, 26, 42–44]; our result therefore further strengthens confidence in its implementation.

Our mechanization is axiom-free and is available at ANONYMIZED (included as supplementary material).

2 Preliminaries: syntax and informal semantics of cut

In the entire paper we use fig. 1 as our running example, and we start by describing how we represent a program like that one. In the concrete syntax, variables are written with capital letters, and predicate application follows the λ-calculus style (no parentheses).

The description of a program $\mathbb{P}$ is shared by both semantics and consists of a list of rules together with a signature environment. We delay the discussion of signatures to section 6 where we describe the static analysis that concerns them and we now focus on the rules.

A rule $\mathbb{R}$ is a pair $\mathbb{T} \times \text{list } \mathbb{A}$, where the first element, a term, represents the rule's head and the second, a list of atoms, represents the rule's premises.

$$\mathbb{T} := \text{Symb } \mathbb{N} \mid \text{Var } \mathbb{N} \mid \text{App } \mathbb{T}\ \mathbb{T}$$
$$\mathbb{A} := \text{cut} \mid \text{call } \mathbb{T}$$

An atom is either the cut operator or a call to a term. Terms include predicate and data symbols, variables, and term application.

The syntax tree allows variables to be applied and predicates to be mixed with data. These higher-order features play no role in the current paper, which focuses on first-order logic programming, but they allow future extensions to full λprolog to be based on the same mechanization.

Variables, predicates and data symbols are identified by natural numbers. We represent substitutions (of type Σ) as finitely supported maps from variables to terms, using the finmap library [31]. The empty substitution is written ε, while the composition of two substitutions σ_1 and σ_2 with disjoint supports is written $\sigma_1 + \sigma_2$.

The backchaining operation, central to both semantics, applies all rules to a query term:

$$\text{backchain} : \mathbb{P} \to \mathcal{F}_v \to \mathbb{T} \to \Sigma \to (\mathcal{F}_v \times \text{list } (\Sigma \times \text{list } \mathbb{A}))$$

Since a rule may be applied multiple times, its variables must be freshened before each application. Accordingly, the backchain function takes a program, the set of variable names used so far ($\mathcal{F}_v$), a query term, and the current substitution. It returns an updated set of variable names together with the list of alternatives, i.e. the rules that apply. More precisely, for each applicable rule it returns the rule premises together with an updated substitution obtained by unifying the rule head with the query term. The unification function takes two terms t_1 and t_2, together with a substitution σ, as input, and optionally returns a substitution σ' that extends σ. If t_1 and t_2 unify starting from σ, then $\sigma'\ t_1 = \sigma'\ t_2$, where $\sigma\ t$ denotes substitution application and $=$ denotes syntactic equality.

To keep the mechanization axiom-free, we implement a unification algorithm, which we briefly describe in section 6 and prove sound and complete. Since the two semantics and the static analysis use the same unification algorithm, the equivalence proof is completely agnostic to the particular implementation chosen for that algorithm.

2.1 The cut operator

There are several variants of the cut operator in the literature. The most widely adopted one is the *hard cut* (see, e.g., the documentation of SWI-Prolog [41]). Whenever the backchain function returns a non-empty list of alternatives, the first is explored immediately, while the others are suspended as choice points. Within a given alternative, premises are processed from left to right, executing each atom in turn. When a cut is encountered, it discards (i) all choice points created by backchaining for the current call and (ii) all choice points created while executing the premises to the left of the cut.

```
func f int -> int.    % f is a deterministic predicate.
f 1 2.                % The two rules have non-verlapping
f 2 3.                % heads, i.e. 1 != 2

func g int -> int.          % g is a deterministic predicate.
g A C :- r A B, f B C, !.   % If the first rule succeeds, the second rule and the choice points left by r are cut
g D F :- f D E, f E F.      % The second rule is the last one and only calls functions

pred r int -> int.    % r is a non-deterministic predicate.
r 0 0.                % The heads overlap: the query for 0
r 0 1.                % has three applicable rules.
r 0 2 :- !.
```

Figure 1. Running example of a logic program in Elpi syntax

For example, consider the program in fig. 1 and the query `g 0 R`. Both rules for the predicate `g` apply to this query. Backchaining therefore returns the following two alternatives:

$$[(\sigma_1, [\texttt{r A B, f B C, !}]),\ (\sigma_2, [\texttt{f D E, f E F}])]$$

with $\sigma_1 := \{\texttt{A} \mapsto \texttt{0}, \texttt{C} \mapsto \texttt{R}\}$ and $\sigma_2 := \{\texttt{D} \mapsto \texttt{0}, \texttt{F} \mapsto \texttt{R}\}$. For simplicity, we choose an example where variable refreshing plays no role, so we do not discuss the set $\mathcal{F}_v$ any further.

Execution continues with the first premise of the first alternative, namely `r A B` under σ_1, i.e. `r 0 B`. The remaining alternatives are stored as choice points. Backchaining `r 0 B` returns $[(\sigma_3, []), (\sigma_4, []), (\sigma_5, [\texttt{!}])]$ where $\sigma_3 := \sigma_1 + \{\texttt{B} \mapsto \texttt{0}\}$, $\sigma_4 := \sigma_1 + \{\texttt{B} \mapsto \texttt{1}\}$ and $\sigma_5 := \sigma_1 + \{\texttt{B} \mapsto \texttt{2}\}$.

The next step of interpretation continues with `f B C` under σ_3, that is, `f 0 R`, and leaves $[(\sigma_4, []), (\sigma_5, [\texttt{!}])]$ as new choice points. This time, backchaining fails (no rule for `f` matches input 0). We therefore backtrack to the most recently introduced choice point and retry `f B C` under σ_4, i.e. `f 1 R`. This succeeds with $\sigma_6 := \sigma_4 + \{\texttt{R} \mapsto \texttt{2}\}$.

We now reach the cut. At this point there are still two unexplored alternatives: one from the third rule for `r` and one from the second rule for `g` (respectively, σ_5 and σ_2). Both are discarded, because they were created when, or after, the first rule for `g` was selected. In contrast, a cut does not discard choice points created earlier; in this example, there are none. The final solution is: $\{\texttt{A} \mapsto \texttt{0}, \texttt{B} \mapsto \texttt{1}, \texttt{C} \mapsto \texttt{R}, \texttt{R} \mapsto \texttt{2}\}$.

In sections 3 and 4, we provide fully mechanized operational semantics for Prolog with cut. They differ in how they represent the ongoing computation, in particular how alternatives are stored and how they are pruned by cut.

Notational conventions In the following section we note $\mathbb{B}$ for the boolean type. Following the Mathematical Components library [29], on which we depend, we use `x.1` and `x.2` to denote the first and second projection applied to the pair `x`, and $l_1 ++ l_2$ for the concatenation of the two lists l_1 and l_2. We use the notation $x \cap y = \varnothing$ to denote that the two sets x and y are disjoint. When the arguments are terms, as in $t_1 \cap t_2 = \varnothing$, we mean that the two terms do not share any variables. Similarly $t \subseteq v$ means that the set of variables occurring in t is included in the set v; $\sigma \subseteq v$ means that v includes the domain of σ and the all variables occurring in its codomain.

3 Stack-based semantics

The stack-based semantics we present is essentially the one by Debray and Mishra in [17, Section 4.3]. The main difference is that we hide the search for applicable rules in the backchain auxiliary function.

The execution state is a stack of alternatives.

$$\text{alts} := \text{list}\ (\Sigma \times \text{goals}) \qquad \text{goals} := \text{list}\ (\mathbb{A} \times \text{alts})$$

Intuitively each stack item is a self-contained computation, coupling a substitution with all the goals to be solved. The datatypes of goals and alternatives are mutually recursive since a goal pairs an atom with a list of the *cut-to* alternatives. Still, these alternatives have a very different role: they have to be understood as pointers to lower levels of the stack itself. They are used to implement the cut, i.e. to prune the stack of alternatives. The run_s inductive predicate describes the stack-based semantics.

$$\text{run}_s : \mathbb{P} \to \mathcal{F}_v \to \text{alts} \to \text{option}\ (\Sigma \times \text{alts}) \to \textbf{Prop}$$

The run_s predicate relates a program, a set of variables and a list of alternatives to an optional result. None stands for failure, i.e., no solution, whereas Some (σ, a) represents success with σ being the solution and a as the list of not-yet-explored alternatives.

The rule system for run_s is given in fig. 2 and consists of four cases. We distinguish two main cases. If we run out of alternatives we fail: rule Fail_s stops the execution and returns

$$\mathcal{B}_S\ p\ v\ t\ \sigma\ a\ g = \text{let}\ (v', rs) := \text{backchain}\ p\ v\ t\ \sigma\ \text{in}\ \left(v', \left[\sigma', ([(q, a) \mid q \in b] ++ g) \;\middle|\; (\sigma', b) \in rs\right]\right)$$

$$\frac{}{\text{run}_s\ p\ v\ ((\sigma,\ [])\ ::\ a)\ (\text{Some}\ (\sigma,\ a))}\ \text{Stop}_s$$

$$\frac{\text{run}_s\ p\ v\ ((\sigma,\ g)\ ::\ \text{ca})\ r}{\text{run}_s\ p\ v\ ((\sigma,\ (\text{cut},\ \text{ca})\ ::\ g)\ ::\ \_)\ r}\ \text{Cut}_s$$

$$\frac{\mathcal{B}_S\ p\ v\ t\ \sigma\ a\ g = (v',\ \text{bs}) \qquad \text{run}_s\ p\ v'\ (\text{bs}\ ++\ a)\ r}{\text{run}_s\ p\ v\ ((\sigma,\ (\text{call}\ t,\ \_)\ ::\ g)\ ::\ a)\ r}\ \text{Call}_s$$

$$\frac{}{\text{run}_s\ \_\ \_\ []\ \text{None}}\ \text{Fail}_s$$

Figure 2. Rule system for run_s

None. If the list of alternatives is $(\sigma_0, h) :: a$, then we look at the list of goals h (under the substitution σ_0). If h is empty, rule Stop_s stops the execution with a success σ_0 leaving a as the unexplored alternatives. If h is not empty we look at its head goal (t, ca) where t is the atom and ca is its cut-to alternatives. If t is a cut, rule Cut_s continues to evaluate the tail of h with alternatives ca. Finally, when t is a call, rule Call_s backchains: for each rule it pairs each premise with the current stack of alternatives, and prepends the obtained goals to the tail of h. Note that if backchain returns the empty list, the second premise of rule Call_s amounts to exploring the next item in the current stack.

3.1 Example of execution

We propose an example of the execution of run_s in fig. 3. The arrows in the images represent the cut-to pointers.

We have not drawn arrows coming out of call atoms since the interpreter ignores the cut-to alternatives in this case. The evolution of the stack in fig. 3 mirrors the example in section 2.1: we evaluate the first goal of the first alternative. During backchaining, we add a pointer to the remaining alternatives to each new cut operator.

If a cut is evaluated, we replace the remaining alternatives with the cut-to alternatives. For example, in fig. 3f, evaluating the cut discards the second and third elements in the stack. Backtracking is performed by simply consuming the first alternative from the stack, as shown in fig. 3d.

4 And-Or Tree semantics

An And-Or tree [25] is a stratified, variable-width tree, defined as follows:

$$\begin{aligned}\mathcal{T} := \quad & \text{Top} \mid \text{Bot} \mid \text{Unexplored } \mathbb{A} \\ & \mid \text{Or } (\text{option } \mathcal{T})\ \Sigma\ \mathcal{T} \mid \text{And } \mathcal{T}\ (\text{list } \mathbb{A})\ \mathcal{T}\end{aligned}$$

It consists of two kinds of layers that alternate: a disjunctive layer is followed by a conjunctive layer and vice versa. At the root (a query), the tree records a list of alternatives (an Or-layer); for each alternative, it records a list of subqueries to be solved (an And-layer). Intuitively, solving a query requires solving one alternative in the Or-layer, but all subqueries in the corresponding And-layer.

The tree is built incrementally. The Unexplored node represents an unexplored branch (an atom). When the atom is a call, we use the backchaining procedure from section 2 to compute the two layers to attach to the tree: alternatives form the Or-layer, and the premises of each applicable rule form the corresponding And-layer. This expansion step is performed by the step procedure described in section 4.2.

In addition to the Unexplored node we have two distinguished nodes, Bot and Top, written $\bot$ and $\top$, which represent the smallest failing and succeeding tree, respectively.

The Or node, written $A \vee B_\sigma$, represents the addition of an alternative B_σ to an existing disjunction A. The left subtree A is optional and is absent once that branch has been fully explored. The right alternative is paired with a substitution σ, which becomes the current substitution when backtracking to B.

The And node, written $A \wedge_{B_0} B$, represents the addition of a subquery B to the first choice point in A. We call B_0 the *reset point* for B: it represents the continuation for all alternatives of A except the first. That is, when backtracking occurs within A, the subtree B is replaced by the atoms in B_0. An example is shown in section 4.3.

4.1 The semantics

The tree is constructed incrementally by two recursive functions, step and prune. Both traverse the tree depth-first, left-to-right.

The type of these two functions is given below.

$$\begin{aligned}\text{step} &: \mathbb{P} \to \mathcal{F}_v \to \Sigma \to \mathcal{T} \to (\mathcal{F}_v \times \text{tag} \times \mathcal{T}) \\ \text{prune} &: \mathbb{B} \to \mathcal{T} \to \text{option } \mathcal{T}\end{aligned}$$

The step routine, among its outputs, also returns a *tag*, whose definition is described in the very following section.

To avoid the constraint that all functions must terminate in Rocq, we define the transitive closure of step and prune as the inductive relation run_t. More precisely run_t iterates calls to step on incomplete trees to expand Unexplored nodes. When a tree fails it calls prune to find the next alternative and continues from there, if one exists. The node to be explored in a tree is retrieved thanks to next_tree (in fig. 4). When this node is Top we say the entire tree is a *success* (short ☑); when it is Bot we say the tree *failed* (short ⊠); otherwise the tree is *incomplete* (short ⊡).

A graphical representation of the execution of the running example on the tree-based semantics is shown in fig. 5. For compactness, we depict And and Or nodes as n-ary, with children associating to the right. The Bot node acts as the terminating element of an Or list, while Top is the terminating element of an And list. We mark with a black arrow the result of next_tree.

4.2 The step procedure

The step procedure takes as input a program, a set of variables, a substitution, and a tree, and returns an updated set of variables, a tag, and an updated tree.

The set of variables is assumed to contain all variables occurring in the tree. It is passed to backchain so that rules can be refreshed correctly.

The tag is implemented as follows:

$$\text{tag} := \text{Expanded} \mid \text{CutBrothers} \mid \text{Failed} \mid \text{Success}$$

It indicates which kind of step was performed. Failure and Success are returned for trees that satisfy, respectively, the failed and success tests. CutBrothers indicates the interpretation of a superficial cut: step was called on an And-layer and its next_tree is the cut atom. Finally, Expanded accounts

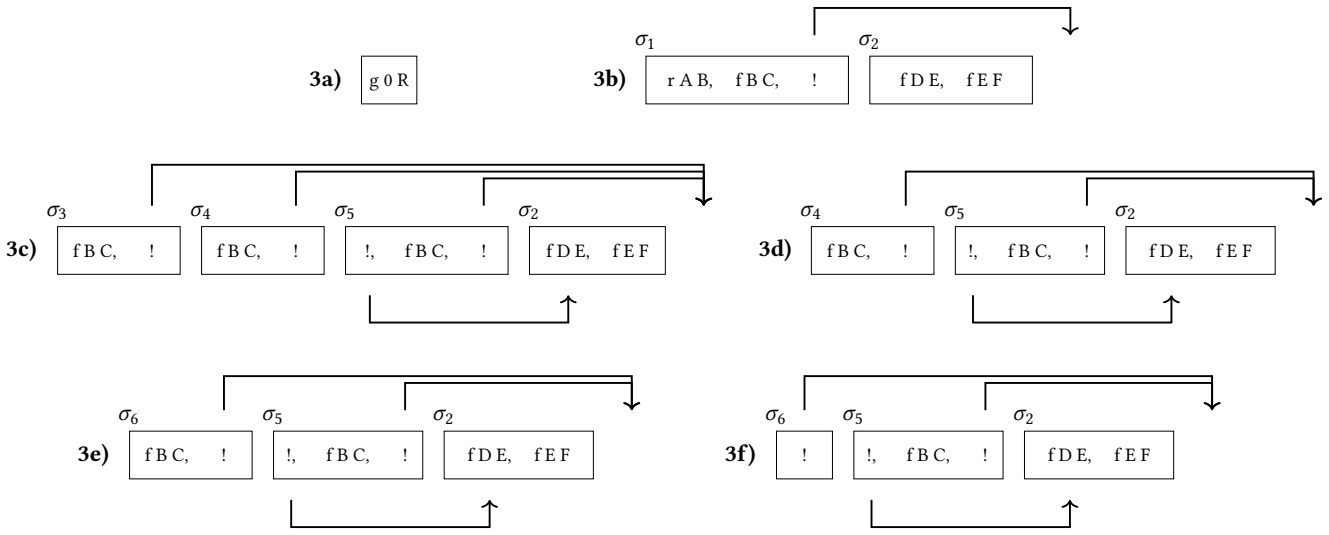

Figure 3. Execution in the stack semantics. The substitutions $\sigma_1 \ldots \sigma_6$ are from section 2.1

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```
Fixpoint next σ t : Σ * tree :=
  match t with
  | Unexplored _ | Bot | Top ⇒ (σ, t)
  | Or None σ' B ⇒ next σ' B
  | Or (Some A) _ _ ⇒ next σ A
  | And A _ B ⇒ let (σ', pA) := next σ A in
                if pA == Top then next σ' B else (σ', pA)
  end.
Definition next_subst σ t := fst (next σ t).
Definition next_tree t := snd (next ε t).
Definition success t := next_tree t == Top.
Definition failed t := next_tree t == Bot.
Definition incomplete t := match next_tree t with
  | Unexplored _ ⇒ true | _ ⇒ false end.
```

Figure 4. The next routine and derived functions

for the interpretation of predicate calls and non-superficial cuts.

The step procedure leaves the tree unchanged when the predicate success (or failed) returns true (i.e. when next_tree is Top or Bot), while it expands Unexplored nodes by calling backchain, which returns a list l of alternatives. If l is empty, then the Unexplored atom is replaced by Bot. Otherwise, it is replaced by a right-skewed tree of Or nodes, where each leaf corresponds to a list of atoms $e \in l$. If e is empty, the leaf is Top; otherwise, it is a right-skewed tree of And nodes.

Fig. 5 depicts the tree corresponding to the running example (section 2.1) as it undergoes calls to step and prune. We run query `g 0 R` against the program P in fig. 1. Let c_1 and c_2 be the children of the root: by construction they correspond respectively to the premises of rules r_1 and r_2 in P. Note that c_1 (resp. c_2) is associated with substitution σ_1 (resp. σ_2), whose values are the same as in section 2.1. Reset points are defined as follows: $R_1 \coloneqq$ [`f B C, !`], $R_2 \coloneqq$ [`!`], and $R_3 \coloneqq$ [`f E F`]. Intuitively, they record the lists of atoms used to generate the corresponding subtree. Other examples of step performing a tree expansion via backchain appear in figs. 5c to 5e.

4.2.1 The interpretation of a cut.

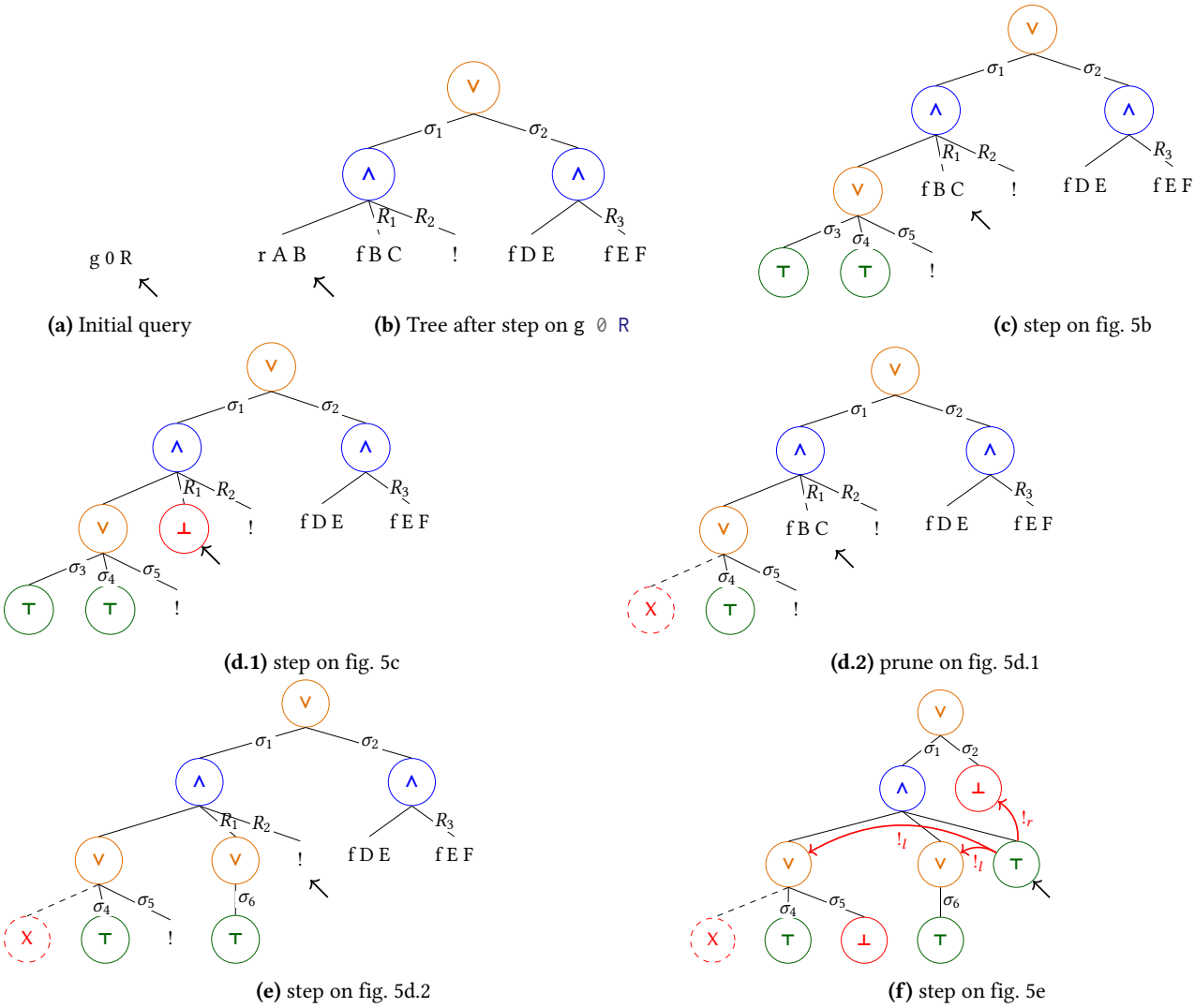
The step corresponding to a cut performs two actions: (i) the cut node is replaced by Top; and (ii) the subtrees in the scope of Cut are pruned. More precisely, (ii) prunes the left siblings of the Cut node (in the same And-layer, denoted $!_l$) and prunes the right uncles of Cut (in the one-level-up Or-layer, denoted $!_r$).

An example of the impact of cut is shown in fig. 5f. The red arrows indicate the scope of the cut and are labeled with $!_r$ and $!_l$ to indicate which pruning operation is applied to each pointed subtree. Note that the evaluation of a cut keeps the path leading to the cut node unchanged, in the sense that substitution σ_4 is preserved, while the path under substitution σ_5 (which also succeeds) is replaced by Bot. This amounts to discarding the third rule of predicate `r`.

4.3 The prune procedure

When backchain returns an empty list, step produces a tree that failed and the computation must backtrack. The prune procedure is responsible for producing the new tree from which the computation can continue.

In fig. 5d.1 we encounter a failure because no rule applies to the atom `f B C` under substitution σ_3, which maps `B` to `0`. The prune procedure prunes the path leading to this failure and resets the current sub-query to its original shape. More precisely, it kills the Top node under σ_3 (since that branch has been fully explored and produced no solution), and then replaces the Bot node with the information stored in the



(a) Initial query **(b)** Tree after step on g `0` R **(c)** step on fig. 5b

(d.1) step on fig. 5c **(d.2)** prune on fig. 5d.1

(e) step on fig. 5d.2 **(f)** step on fig. 5e

Figure 5. Execution in the tree semantics. The substitutions $\sigma_1 \ldots \sigma_6$ are from section 2.1. R_1, R_2, and R_3 are reset points and correspond respectively to the lists of atoms [`f B C, !`], [`!`], and [`f E F`].

reset point R_1. This restores the call `f B C`, which can then be reconsidered under substitution σ_4.

Furthermore, prune serves another important purpose. Its behavior depends on the boolean flag it receives as input: prune false t recursively removes all failures (if any) from t, whereas prune true t removes the first success in t. For example, the tree in fig. 5f contains a successful path that maps `R` to `2`. Calling prune on this tree returns None, since there are no alternatives in the tree after the success.

4.4 The run_t relation and its properties

The inductive relation run_t has the following type:

$$\text{run}_t : \mathbb{P} \to \mathcal{F}_v \to \Sigma \to \mathbb{T} \to \text{sol} \to \mathbb{B} \to \mathcal{F}_v \to \texttt{Prop}$$

It associates a program, a set of variables, an initial substitution σ, and a tree t with a solution r, a boolean b, and an updated set of variable names v'.

The type of sol is defined as follows:

$$\text{sol} := \text{Zero} \mid \text{One}\ \Sigma \mid \text{Many}\ \Sigma\ \mathcal{T}$$

Zero represents a failing execution. One σ represents an execution producing exactly one solution, i.e. no backtracking is possible from the resulting state. It also carries the substitution σ that represents the solution. Finally, Many σ t represents an execution in which the original state produces the solution σ and leaves the alternatives in t unexplored.

The boolean b records whether a superficial cut was evaluated during execution, and v' is the updated set of variables tracking freshly generated names.

$$\frac{\boxtimes\ t \quad \text{next\_subst}\ \sigma\ t \rightsquigarrow \sigma' \qquad \text{prune true}\ t \rightsquigarrow \text{None}}{\text{run}_t\ p\ v\ \sigma\ t \rightsquigarrow (\text{One}\ \sigma', \text{false}, v)}\ \text{Stop}_t^o$$

$$\frac{\boxtimes\ t \quad \text{next\_subst}\ \sigma\ t \rightsquigarrow \sigma' \quad \text{prune true}\ t \rightsquigarrow \text{Some}\ t'}{\text{run}_t\ p\ v\ \sigma\ t \rightsquigarrow (\text{Many}\ (\sigma', t'), \text{false}, v)}\ \text{Stop}_t^m \qquad \frac{\boxdot\ t \quad \text{step}\ p\ v\ \sigma\ t \rightsquigarrow (v', tg, t') \quad \text{run}_t\ p\ v'\ \sigma\ t' \rightsquigarrow (r, b, v'')}{\text{run}_t\ p\ v\ \sigma\ t\ \rightsquigarrow (r, (tg = \text{CutBrothers}) \vee b, v'')}\ \text{Step}_t$$

$$\frac{\boxtimes\ t \quad \text{prune false}\ t \rightsquigarrow \text{Some}\ t' \quad \text{run}_t\ p\ v\ \sigma\ t' \rightsquigarrow r}{\text{run}_t\ p\ v\ \sigma\ t \rightsquigarrow r}\ \text{Back}_t \qquad \frac{\text{prune false}\ t \rightsquigarrow \text{None}}{\text{run}_t\ p\ v\ \sigma\ t \rightsquigarrow (\text{None}, \text{false}, v)}\ \text{Fail}_t$$

Figure 6a. Rule system for run_t

$$\frac{A \quad (A\ \text{cuts})}{A \vee B}\ (\vee_{il}) \qquad \frac{\neg A \quad (A\ \text{cuts})}{\neg(A \vee B)}\ (\vee_{ir1}) \qquad \frac{\neg A \quad B \quad (A\ \text{does not cut})}{A \vee B}\ (\vee_{ir2})$$

$$\frac{A \vee B \qquad \overset{[A]}{C} \qquad (A\ \text{cuts})}{C}\ (\vee_{e1}) \qquad \frac{A \vee B \qquad \overset{[A]}{C} \qquad \overset{[\neg A, B]}{C} \quad (A\ \text{does not cut})}{C}\ (\vee_{e2})$$

Figure 6b. Introduction and elimination rules for disjunction (Theorems 4.1 to 4.5)

The run_t predicate has five cases, depicted as inference rules in fig. 6a.

Rules Stop_t^o and Stop_t^m say that when the tree t is a success, the substitution σ' is extracted from t (starting from σ); depending on the result of prune true t – which cleans the tree of its successful path – we return One or Many, according to whether alternatives remain to be explored. Rule Step_t applies when next_tree t is an atom: step is invoked to evaluate t, and run_t is called recursively on the updated tree. When next_tree t is a failure, rule Back_t calls prune to clear the failed path; if the resulting cleaned tree exists, run_t is called recursively on it. Finally, rule Fail_t accounts for trees with no solutions.

4.4.1 Properties of run_t.

The boolean output of run_t allows us to state interesting properties about the Or node in the presence of cut (fig. 6b). To simplify their presentation, we introduce the following notation for Or nodes. When the left-hand side is None, we write the symbol $\square$. Otherwise, the notation $A \vee B_\sigma$ implicitly denotes $(\text{Some}\ A) \vee B_\sigma$.

Theorem 4.1 (or_intro_left). $\forall\ p\ v\ v'\ \sigma\ \sigma'\ A\ A'\ \sigma_1\ B$,
$\text{run}_t\ p\ v\ \sigma\ A\ (\text{Many}\ \sigma'\ A')\ \text{true}\ v' \rightarrow$
$\text{run}_t\ p\ v\ \sigma\ (A \vee B_{\sigma_1})\ (\text{Many}\ \sigma'\ (A' \vee \text{Bot}_{\sigma_1}))\ \text{false}\ v'$

Theorem 4.2 (or_intro_right). $\forall\ p\ v\ v'\ \sigma\ A\ \sigma_1\ B$,
$\text{run}_t\ p\ v\ \sigma\ A\ \text{Zero true}\ v' \rightarrow \text{run}_t\ p\ v\ \sigma\ (A \vee B_{\sigma_1})\ \text{Zero false}\ v'$

Theorem 4.3 (or_intro_right2). $\forall\ p\ v_0\ v_1\ v_2\ \sigma\ A\ \sigma_1\ \sigma_2\ B\ b$,
$\text{run}_t\ p\ v_0\ \sigma\ A\ \text{Zero false}\ v_1 \wedge \text{run}_t\ p\ v_1\ \sigma_1\ B\ (\text{One}\ \sigma_2)\ b\ v_2 \rightarrow$
$\text{run}_t\ p\ v_0\ \sigma\ (A \vee B_{\sigma_1})\ (\text{One}\ \sigma_2)\ \text{false}\ v_2$

Theorems 4.1 to 4.3 correspond to the introduction rules for disjunction. If A executes a cut, one *cannot* obtain $A \vee B$ from B: the execution of A, whether it ultimately succeeds or fails, makes the success of B irrelevant (the cut discards it). In the first lemma, B is forced to be Bot; in the second one, the failure of A absorbs B. The last lemma is connected to the depth-first exploration of the tree: proving a disjunction from its right-hand side can only occur after the left-hand side has been disproved.

Theorems 4.4 and 4.5 are elimination rules for disjunction:

Theorem 4.4 (or_elim1). $\forall\ p\ v\ v'\ \sigma\ \sigma'\ \sigma_1\ A\ A'\ B\ B'$,
$\text{run}_t\ p\ v\ \sigma\ (A \vee B_{\sigma_1})\ (\text{Many}\ \sigma'\ (A' \vee B'_{\sigma_1}))\ \text{false}\ v' \rightarrow$
$\exists\ b,\ \text{run}_t\ p\ v\ \sigma\ A\ (\text{Many}\ \sigma'\ A')\ b\ v' \wedge B' = \text{if}\ b\ \text{then}\ \bot\ \text{else}\ B$

Theorem 4.5 (or_elim2). $\forall\ p\ v\ v'\ \sigma\ \sigma'\ \sigma_1\ A\ B\ B'$,
$\text{run}_t\ p\ v\ \sigma\ (A \vee B_{\sigma_1})\ (\text{Many}\ \sigma'\ (\square\ \vee B'_{\sigma_1}))\ \text{false}\ v' \rightarrow$
$(\text{run}_t\ p\ v\ \sigma\ A\ (\text{One}\ \sigma')\ \text{false}\ v') \vee$
$(\exists\ v_2\ b,\ \text{run}_t\ p\ v\ \sigma\ A\ \text{Zero false}\ v_2 \wedge$
$\text{run}_t\ p\ v_2\ \sigma_1\ B\ (\text{Many}\ \sigma'\ B')\ b\ v')$

From $A \vee B$, when A is not inert (i.e. not already explored), we cannot deduce A or B independently. Instead, we can only derive a weakened conclusion of the form $\top \vee B'$, where B' provides no information in the case where the exploration of A performs a cut. This yields an elimination rule that is stronger than the usual one. If A does not cut, the elimination rule has the usual two premises, together with the additional constraint that whenever B holds, A must have failed. This again reflects the depth-first traversal of the proof tree.

It is worth recalling that the "A cuts" side-condition in the rules above is precisely the boolean value returned by run_t. Without this information, it is impossible to formulate these rules: it is not merely a matter of whether A contains a syntactic occurrence of cut, but whether that cut is actually executed during the (dis)proof of A (in rules $(\vee_{ir1})$ and $(\vee_{ir2})$). Indeed, an atom preceding the cut may fail, so the cut is never reached.

As anticipated in the introduction, these lemmas show that, in the presence of cut, disjunction is not commutative.

4.5 Inadequacy of the stack-based semantics for formal reasoning

The lemmas that we presented in section 4.4.1 are hard to express in the stack-based semantics, due to the lack of structure in the stack to delimit the scope of the cut operator. For example, consider a stack $a_0, a_1, \ldots, a_n$. If a_0 succeeds but executes a cut, it is hard to relate the remaining alternatives

$a_1, \ldots, a_n$ to the alternatives $b_1, \ldots, b_m$ that survive the cut, since the only simple invariant is that $b_1, \ldots, b_m$ is a suffix of $a_1, \ldots, a_n$. If $a_1, \ldots, a_n$ were generated before the call whose execution generates the cut in a_0, then they all stay; but if some were generated afterwards, they are discarded. Expressing this precisely would require attaching, at the very least, time-stamps (or depths) to goals, which amounts to a cumbersome encoding of a tree.

5 Equivalence of the two semantics

Before proving the equivalence of the two semantics, we first need to relate trees to stacks.

5.1 From trees to stacks

The function t2s takes as input the tree t_0, a substitution σ for the input tree, and an auxiliary list of choice points cp.

$$\text{t2s} : \mathcal{T} \to \Sigma \to \text{alts} \to \text{alts}$$

These choice points represent alternatives that appear at the same level of the current tree, such as the right siblings of an Or node.

The Top node represents one successful alternative; therefore it is translated into a singleton list whose only element has an empty list of goals. The Bot node represents failure, hence it is mapped to the empty list of alternatives. Finally, atoms are mapped to a singleton list containing one alternative whose only goal is that atom, with an empty cut-to list. At this stage, the cut-to list cannot point to cp: atoms are not right-uncle subtrees, whereas cp represents alternatives that will actually be discarded if the atom turns out to be a cut. The correct cut-to list is therefore determined later, once the trees corresponding to sibling subtrees have been inspected.

The translation of $A \vee B_\sigma$ creates two lists of alternatives, one for A and the other for B. The alternatives of B are valid choice points for A and are therefore passed to the translation of A. The two lists of alternatives can then be concatenated. Moreover, every atom occurring one level below cp (i.e. in A or B) must add cp to its cut-to alternatives.

The translation of $A \wedge_{B_0} B$ starts with the translation of A. If the translation of A is an empty list, we return the empty list of alternatives without even inspecting B or B_0, since an empty translation of A indicates failure, which in turn implies the failure of the entire conjunction. When the translation of A is a list $(\sigma_0, g) :: a$, the B node represents the continuation of the first alternative g, whereas B_0 is the continuation for each of the remaining alternatives in a.

The function t2s is not injective: distinct trees can translate to the same stack. In particular, a transition from a tree t_1 to a tree t_2 performed by run_t may leave both trees mapped to the same stack. We refer to such a transition as a *silent transition*.

Notably, the stack semantics has no notion of a failed stack. Therefore, when proving the equivalence between the two semantics, we must take into account that run_t may perform more iterations than run_s before returning a solution or reporting a failure. We illustrate this non-injective behavior with a concrete example below.

5.1.1 Example of t2s execution.

The translations of the trees in fig. 5 are shown in fig. 3. The letters in the figures relate each tree to the corresponding stack. For example, 5b is translated into 3b. We also note that 5d.1 and 5d.2 are both translated into 3d, confirming that t2s is not injective: different trees can map to the same stack. Consequently, some transitions performed by run_t are not visible in run_s. As an example of a call to t2s, we consider the tree and the stack in fig. 7, taken respectively from figs. 3c and 5c. The red arrows point from the head of an alternative in the tree to the head of the corresponding stack cell. We focus on the deepest Or node in the tree. This subtree is a disjunction with four children. The first is ignored, since it is None. The success node below σ_3 has `f B C` and the cut operator as continuation, corresponding to the first stack cell. We note that the cut operator points outside the stack, since the scope of this cut eliminates its right uncle. The success node below σ_4 has R_1 as continuation, where R_1 is the list of goals `f B C, !`; this is translated into the second cell of the stack. Finally, the cut operator under σ_5 also has R_1 as continuation; it corresponds to the third alternative. We can see that the first cut points to the translation of the subtree under σ_2, since that subtree is one Or-layer above its right uncles.

5.2 Equivalence

The equivalence we are interested in states that a query produces the same substitution and alternatives under both semantics. The tree-based semantics exposes more information than this comparison requires, so we hide the surplus behind existential quantifiers.

Definition 5.1 (run'_t). $\lambda\ p\ v\ \sigma\ t\ r.$
$\exists\ b\ v',\ \text{run}_\text{t}\ p\ v\ \sigma\ (\text{Unexplored}\ t)\ r\ b\ v'$

For each possible outcome, we prove the equivalence of the two semantics.

Corollary 5.2 (equiv_zero). $\forall\ p\ t\ v,\ t \subseteq v \to$
$\text{run}_\text{s}\ p\ v\ [\varepsilon,\ [\text{call}\ t,\ []]]\ \text{None} \leftrightarrow$
$\text{run}'_\text{t}\ p\ v\ \varepsilon\ (\text{call}\ t)\ \text{Zero}$

Corollary 5.3 (equiv_one). $\forall\ p\ t\ \sigma'\ v,\ t \subseteq v \to$
$\text{run}_\text{s}\ p\ v\ [\varepsilon,\ [\text{call}\ t,\ []]]\ (\text{Some}\ (\sigma',\ []))\ \leftrightarrow$
$\text{run}'_\text{t}\ p\ v\ \varepsilon\ (\text{call}\ t)\ (\text{One}\ \sigma')$

Corollary 5.4 (sound_many). $\forall\ p\ t\sigma'\ t'\ v,\ t \subseteq v \to$
$\text{run}'_\text{t}\ p\ v\ \varepsilon\ (\text{call}\ t)\ (\text{Many}\ \sigma'\ t') \to$
$\text{run}_\text{s}\ p\ v\ [\varepsilon,\ [\text{call}\ t,\ []]]\ (\text{Some}\ (\sigma',\ \text{t2s}\ t'\ \sigma\ []))$

Corollary 5.5 (complete_many). $\forall\ p\ t\ \sigma'\ x\ \text{xs}\ v,\ t \subseteq v \to$
$\text{run}_\text{s}\ p\ v\ [\varepsilon,\ [\text{call}\ t,\ []]]\ (\text{Some}\ (\sigma',\ x :: \text{xs})) \to$
$\exists\ t',\ \text{run}'_\text{t}\ p\ v\ \varepsilon\ (\text{call}\ t)\ (\text{Many}\ \sigma'\ t') \wedge \text{t2s}\ t'\ \varepsilon\ [] = x :: \text{xs}$

Because both semantics rely on the same notion of substitution, corollaries 5.2 and 5.3 can be stated without relating

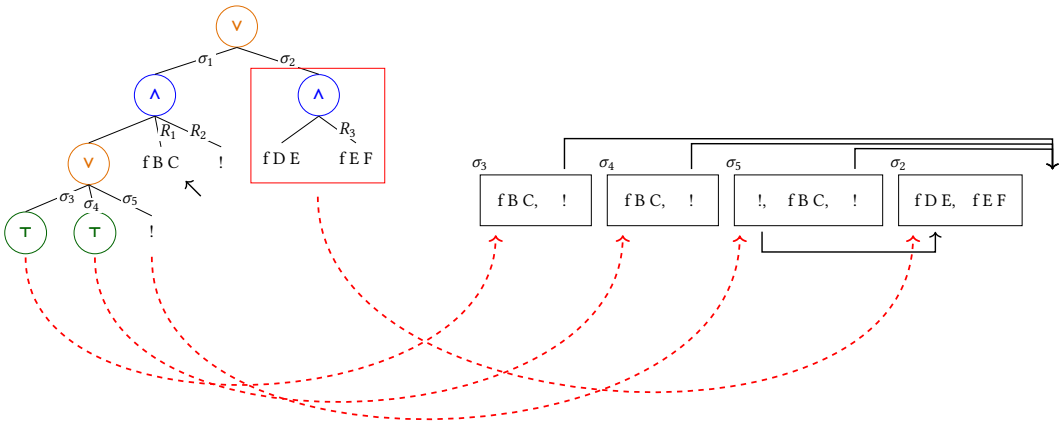


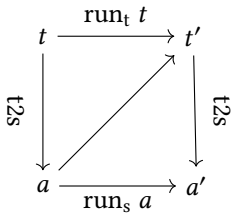
Figure 7. Example of t2s execution

the states — the stacks and trees — on which the two semantics operate. In contrast, in corollaries 5.4 and 5.5 the execution leaves choice points open, so we do need to relate the two kinds of states.

Following the proof technique of [5], we only need the mapping t2s from trees to stacks, already described in section 5.1; the converse mapping is never constructed explicitly. The corollaries above follow from the two theorems proved in section 5.3.

5.3 Equivalence proof

The two theorems below each proceed by induction on the execution of one semantics — the top or the bottom horizontal arrow in the diagram — to derive the other. We begin with the top-to-bottom direction, from trees to stacks.



Since not every tree or stack is reachable from the initial state under either semantics, it is important to identify the subset of valid trees. We defer the exact definition of sld_tree to section 5.3.1.

Theorem 5.6 (runT2S). $\forall\ p\ t\ \sigma\ r\ v,\ t\ \cup\ \sigma \subseteq v \rightarrow$
$\text{sld\_tree}\ t \wedge (\exists\ b\ v',\ \text{run}_t\ p\ v\ \sigma\ t\ r\ b\ v') \rightarrow$

$$\begin{cases} \text{run}_s\ p\ v\ (\text{t2s}\ t\ \sigma\ [])\ \text{None} & \text{if } r = \text{Zero} \\ \text{run}_s\ p\ v\ (\text{t2s}\ t\ \sigma\ [])\ (\text{Some}\ (\sigma', [])) & \text{if } r = \text{One}\ \sigma' \\ \text{run}_s\ p\ v\ (\text{t2s}\ t\ \sigma\ [])\ (\text{Some}\ (\sigma',\ \text{t2s}\ t'\ \sigma\ [])) & \text{if } r = \text{Many}\ \sigma'\ t' \end{cases}$$

Proof. By induction on run_t. □

The converse direction could in principle be proved symmetrically, by defining a mapping from stacks to trees together with a predicate identifying the valid stacks that the semantics can generate from the initial one. The backward simulation technique of [5] that we follow makes both definitions unnecessary, by using an existential quantifier instead (on t' in the third case). Validity of the stack a is recovered for free, since it is obtained from a valid SLD tree; and by inhabiting the existential in a constructive logic, we implicitly obtain the mapping from stacks to trees.

Theorem 5.7 (runS2T). $\forall\ p\ v\ a\ r\ v\ \sigma\ t,\ t\ \cup\ \sigma \subseteq v \rightarrow$
$\text{run}_s\ p\ v\ a\ r \rightarrow\ \text{sld\_tree}\ t \rightarrow \text{t2s}\ t\ \sigma\ [] = a \rightarrow \exists\ b\ v',$

$$\begin{cases} \text{run}_t\ p\ v\ \sigma\ t\ \text{Zero}\ b\ v' & \text{if } r = \text{None} \\ \text{run}_t\ p\ v\ \sigma\ t\ (\text{One}\ \sigma')\ b\ v' & \text{if } r = \text{Some}\ (\sigma',\ []) \\ \exists\ t',\ \text{run}_t\ p\ v\ \sigma\ t\ (\text{Many}\ \sigma'\ t')\ b\ v' & \text{if } r = \text{Some}\ (\sigma',\ a') \\ \quad \wedge\ \text{t2s}\ t'\ \sigma\ [] = a' \end{cases}$$

Proof. By induction on run_s accounting for silent transitions as we align its iterations with run_t. □

5.3.1 Valid SLD trees.

The class of valid SLD trees is captured by the sld_tree function shown in fig. 8. While all base cases of tree are valid, we need to analyze the Or and And constructors more carefully.

For the Or constructor, we distinguish two cases depending on whether the left-hand side is present. If it is not, the right-hand side must be a valid SLD tree. Otherwise, the left-hand side must be a valid SLD tree, and the right-hand side must either be the Bot tree or be unexplored. The former case arises when a superficial cut in the left-hand side cuts away the right-hand side. In the latter case, we use the *base_or* predicate to check that B is the right-skewed tree

```
Fixpoint sld_tree A :=
  match A with
  | Unexplored _ | Top | Bot ⇒ true
  | Or None _ B ⇒ sld_tree B
  | Or (Some A) _ B ⇒
      sld_tree A && ((B == Bot) || base_or B)
  | And A B0 B ⇒
      sld_tree A && if success A then sld_tree B
                                 else B == big_and B0
  end.
```

Figure 8. The sld_tree predicate

formed by a disjunction of conjunctions generated after a successful backchain for a call.

For the And constructor, the left-hand side is required to be a valid SLD tree. The shape of the right-hand side depends on whether the left-hand side has succeeded. If it has not, the right-hand side must not be explored, i.e. B must be the right-skewed tree containing conjunctions of premise atoms. This is enforced by the *big_and* procedure, which generates a tree from the reset point B_0 (the same procedure used to transform each alternative output by backchain into the And-layer of the resulting tree). When the left-hand side has succeeded, the right-hand side may have been explored, and must therefore be a valid SLD tree.

In the mechanization we show that sld_tree is preserved by both step and prune.

6 Case study: determinacy analysis

As an application of the tree semantics, we mechanize the soundness proof of the determinacy-checking algorithm introduced in version 3 of the Elpi language [1]. The static checker takes as input the signatures of the predicates declared by the user and verifies that the rules of a deterministic predicate generate at most one solution; we call this kind of predicate a function. Thanks to this static verification, the user can better control unexpected backtracking.

A predicate behaves deterministically if two conditions are satisfied: *mutual exclusion* guarantees that at most one rule can be successfully applied to any given query, whereas *rule-body checking* ensures that each rule does not create choice points, for example by calling non-deterministic predicates.

6.1 Unification and matching

Our mechanization implements the unification algorithm by Martelli and Montanari [30]; we prove it terminating (using the standard lexicographic ordering), correct, and complete.

Definition 6.1 (idempotent [35]).
$\lambda\sigma.\text{dom}\ \sigma \cap \text{codom}\ \sigma = \varnothing$

Note that idempotent substitutions are necessarily self-applied (hence acyclic), i.e. $\sigma := \{X \mapsto f\ Y, Y \mapsto a\}$ is not idempotent, whereas the equivalent $\sigma' := \{X \mapsto f\ a, Y \mapsto a\}$ is. We prove that the algorithm by Martelli and Montanari generates substitutions that are idempotent.

Definition 6.2 (mgu). $\lambda\ \sigma\ t_1\ t_2.$
$\sigma\ t_1 = \sigma\ t_2 \wedge (\forall \sigma', \text{idempotent}\ \sigma' \wedge \sigma'\ t_1 = \sigma'\ t_2 \longrightarrow$
$\exists \rho, \text{idempotent}\ \rho \wedge \forall t, \rho(\sigma\ t) = \sigma'\ t)$

Lemma 6.3 (unify_correct). $\forall\ t_1\ t_2\ \sigma,$
$\text{unify}\ t_1\ t_2\ \varepsilon = \text{Some}\ \sigma \longrightarrow \text{mgu}\ \sigma\ t_1\ t_2$

Lemma 6.4 (unify_complete). $\forall\ t_1\ t_2,$
$(\exists\ \sigma,\ \text{idempotent}\ \sigma \wedge\ \sigma\ t_1 = \sigma\ t_2) \longrightarrow$
$\exists\ \sigma',\ \text{unify}\ t_1\ t_2\ \varepsilon = \text{Some}\ \sigma'$

The Elpi language adopts a restricted unification routine for input arguments, called matching [4, 46]. Like unify, matching takes a query term t_1 and a pattern t_2, but it also takes a set of frozen variables v and returns an optional substitution σ' that does not assign any variable in v.

To address this need, we extend the Martelli-Montanari unification procedure so that it takes a set of frozen variables. The function matching has the following property:

Lemma 6.5 (matching_disj). $\forall\ v\ t_1\ t_2\ \sigma\ \sigma',$
$\text{idempotent}\ \sigma \wedge \sigma\ t_2\ \subseteq\ v \longrightarrow$
$\text{matching}\ v\ t_1\ t_2\ \sigma = \text{Some}\ \sigma' \longrightarrow \sigma'\ t_1 = \sigma\ t_2$

The backchain function uses matching or unify on the arguments q depending on whether the argument is in *input* or *output* mode.

For the static checker, and in particular to validate rule mutual exclusion, we need to ensure that no two rules can be used on the same query in the absence of cut. We exploit the behavior of matching on input arguments to guarantee this property: mutual exclusion ensures that every pair of rules for a deterministic predicate has an input term that does not unify.

An important property we derive from the matching and unification procedures is the following. Given three pairwise variable-disjoint terms h_1, h_2, and q, representing two rule heads in the database and a query, if both heads match the query q, then the two heads must unify.

Definition 6.6 (opt2prop).
$\lambda x.\text{if}\ x\ \text{is Some \_ then True else False}$

In the following 2 lemmas, we use definition 6.6 as an implicit coercion to cast result of calls to matching and unify into the sort of propositions.

Lemma 6.7 (matching_unify_trans). $\forall\ v\ h_1\ h_2\ q,$
$h_1 \cap h_2 = \varnothing \wedge h_1 \cap q = \varnothing \wedge h_2 \cap q = \varnothing \wedge q \subseteq v \longrightarrow$
$\text{matching}\ v\ h_1\ q\ \varnothing \wedge \text{matching}\ v\ h_2\ q\ \varnothing \longrightarrow \text{unify}\ h_1\ h_2\ \varnothing$

Finally, we also need to prove that the refreshing of variables performed at runtime does not affect unification. In particular, a renaming f for a term t is a function from variables to variables such that: (i) f is defined on all variables in t, i.e. every variable is renamed; and (ii) f is injective, i.e. it does not identify distinct variables. Refreshing therefore produces α-equivalent terms.

Definition 6.8 ($=_\alpha$). $\lambda t_1\ t_2.$
$\exists (r : \text{renaming\_for}\ t_2), t_1 = \text{rename}\ r\ t_2$

Lemma 6.9 (unif_ren_ac). $\forall\ t_1\ t_2\ t_1'\ t_2',$
$t_1 =_\alpha t_1' \wedge\ t_2 =_\alpha t_2' \wedge t_1 \cap t_2 = \varnothing \wedge\ t_1' \cap t_2' = \varnothing \longrightarrow$
$\text{unify}\ t_1\ t_2\ \varepsilon \longrightarrow \text{unify}\ t_1'\ t_2'\ \varepsilon$

6.2 Soundness of determinacy checking

In fig. 1, the rules of the program are equipped with three signatures, one per predicate. Besides the arity of each predicate, a signature specifies which arguments are inputs and

```
Fixpoint det_tree (p: program) A :=
  match A with
  | Unexplored cut => true
  | Unexplored (call t) => tm_is_det p t
  | Bot | Top => true
  | And A B0 B =>
      det_tree p B &&
      ((prune (success A) A == None) ||
       ((det_tree p A || (has_cut B && has_cuts B0)) &&
        det_trees p B0))
  | Or None _ B => det_tree p B
  | Or (Some A) _ B =>
      det_tree p A &&
      if has_cut A then det_tree p B else (B == Bot)
  end.
```

Figure 9. The *det_tree* predicate

which are outputs, their types, and whether the predicate is deterministic. The signatures indicate that `f` and `g` are expected to be functions, as specified by the **`func`** keyword, whereas `r` is a relation, as indicated by the **`pred`** keyword. The **`->`** symbol separates inputs from outputs; therefore, all three predicates are expected to take an `int` as input and return an `int` as output.

The signatures of the program are stored in the `sig` field of the program and are encoded as described in [1]: we distinguish data from predicates, and predicates are classified as deterministic or non-deterministic.

We say that an execution behaves deterministically when, whenever it terminates, it either fails or succeeds leaving no alternative to explore.

Definition 6.10 (is_det). $\lambda\ p\ \sigma\ t,\ \forall\ r\ v,\ t \cup \sigma \subseteq v \longrightarrow \text{run}_s\ p\ v\ [\sigma,\ [t, []]]\ r \longrightarrow r = \text{None} \vee \exists\ \sigma',\ r = \text{Some}\ (\sigma',\ [])$

Due to the inadequacy of the stack semantics to reason about cut, we state the main theorem in terms of the tree semantics and transfer it back using the equivalence proof from section 5.2.

We establish an invariant characterizing deterministic computations.

Definition 6.11 (det_tree). The det_tree predicate, shown in fig. 9, identifies trees whose interpretation does not generate choice points.

For a call to a term t, the tree is deterministic whenever t calls a deterministic predicate. The recursive cases correspond to the And and Or nodes.

A conjunction behaves deterministically whenever its right-hand side is a deterministic tree. Two cases must then be considered. If prune (success A) A = None, then A produces no choice points: it either always fails or contains a single successful execution path. Since det_tree p B holds, the entire conjunction is therefore deterministic. Otherwise, the left-hand side may introduce backtracking, potentially resuming the reset point, which must itself behave deterministically. In this case, either A is deterministic, or both B and the reset point contain a cut that prevents the non-deterministic behaviour of A from arising.

For Or trees, the interesting case is when the left-hand side is present. In this case, A must be deterministic. If A contains a superficial cut, then B must also be deterministic. Indeed, if A succeeds, the cut discards B; otherwise, if A fails before reaching the cut, then no solution is produced by A and B is evaluated.

Finally, if A does not contain a cut, then B must be equal to $\bot$, that is, a tree already discarded by a cut. This condition is essential. Otherwise, if A succeeds without a superficial cut and B is not $\bot$, then B could still produce a successful branch, leading to non-deterministic behaviour.

Definition 6.12 (det_checked). A program is det_checked if all its rules satisfy mutual exclusion and rule-body checking as per [1].

Lemma 6.13 (det_tree_step). $\forall$ p $v\ \sigma\ t\ r,\ t \cup \sigma \subseteq v \longrightarrow$ det_checked p $\wedge$ step u p $v\ \sigma\ t = r \longrightarrow$ det_tree p (snd r)

Lemma 6.14 (det_tree_prune). $\forall\ p\ t\ t'\ b,$ det_tree $p\ t \wedge$ prune $b\ t$ = Some $t' \longrightarrow$ det_tree $p\ t'$

Lemma 6.15 (det_check_tree). $\forall\ \sigma\ v\ p\ t\ r\ b\ v',$ $t \cup \sigma \subseteq v \wedge$ det_checked $p \wedge$ det_tree $p\ t \longrightarrow$ $\text{run}_t\ p\ v\ \sigma\ t\ r\ b\ v' \longrightarrow r = \text{Zero} \vee \exists\ \sigma,\ r = (\text{One}\ \sigma)$

Proof. By induction on run_t using the fact that step and prune preserve det_tree lemmas 6.13 and 6.14. □

Definition 6.16 (tm_is_det). Given a program p and a term t, we say that t is deterministic if its head is a predicate f which is declared as deterministic in p.

Theorem 6.17 (det_check_call). $\forall\ p\ \sigma\ t,$ det_checked $p \wedge$ tm_is_det $p\ t \longrightarrow$ is_det $p\ \sigma$ (call t)

Proof. From lemma 6.15 we conclude that a call to a deterministic predicate behaves deterministically on trees; then, thanks to corollary 5.4, we derive that it also behaves deterministically when executed in run_s. □

7 Related work

The closest work to our tree-based semantics is arguably that of Börger and Rosenzweig [8, 9]. Their computation state is also a tree, but one that branches only on clause alternatives: the goals still to be solved are kept as a flat sequence inside each node, as in an SLD-tree, and cut is implemented by tagging every goal with a *cutpoint* pointer to the node control returns to. Our trees branch on premises as well as on alternatives, so the scope of a cut is recovered from the position of the cut node and the $!_l$ / $!_r$ pruning performed by step needs no cutpoint pointer. Such a pointer is similar in spirit to the cut-to alternatives of the stack-based semantics, and shares their shortcomings for reasoning about

programs that use cut. In any case, the model of [9] is defined on paper and serves as the first model of a refinement chain towards the Warren abstract machine [10]; that line of work targets compiler correctness and does not reason about the behaviour of individual programs.

Closest to our case study is Mogensen [34], who derives a determinacy analysis for Prolog with cut as an abstraction of a denotational semantics, thereby giving a semantic justification to an earlier analysis of Sahlin [38]. As with the boolean returned by run_t, his abstract domain records whether a cut is guaranteed to have been executed. His soundness argument is carried out on paper and the analysis is not connected to a stack- or machine-level semantics.

The only mechanization of an operational semantics for Prolog with cut that we are aware of is Pusch's, in Isabelle/HOL [36]. Starting from the model of [9], she mechanizes a chain of refinements towards the Warren abstract machine [4, 45]. Unlike our case study, she does not use the semantics to prove program properties, and so does not run into the difficulty that led us to a semantics in which the effect of cut is explicit. The two developments are nonetheless complementary: together they suggest that our tree-based semantics is equivalent to an optimized stack-based one, though a fully machine-checked proof is out of reach, since the two are carried out in different provers.

Another relevant mechanization is that of Kriener and King [27], whose Rocq source code we could not locate. Their semantics operates on programs in super-homogeneous form, which would have sufficed for the first-order fragment considered in this paper but cannot easily accommodate all the features covered in [1], in particular dynamic predicates. More importantly, their mechanization does not relate the denotational semantics to an operational semantics of the kind underlying standard Prolog abstract machines, a link that our equivalence proof provides.

A further relevant mechanization is that of Rozplokhas et al. [37], who mechanize in Rocq a semantics that tracks cut. Since they consider only green cuts [37, Appendix C], it cannot serve as a basis for mechanizing [1].

Hanus and Prott [23] formalize in Rocq a determinacy checker for the logic-functional language Curry. There, choice points are denoted explicitly by the `?` node in the syntax of expressions, and the overlap between rules is likewise made syntactic through a program transformation [2]; the choice-point-absorbing operator `allValues` takes the expression whose choice points it discards as an explicit argument, so the scope of this control operator is immediate. Cut has no such syntactic marker: its scope is implicit in the surrounding stack or tree structure, which is precisely what makes relating the two semantics nontrivial in our setting.

The technique we use to relate the two semantics is inspired by Bertot's mechanized proof of compiler correctness [5], which relates an imperative language to its compiled assembly code. That proof uses the compiler as the translation function in one direction and introduces no decompiler, just as we translate trees to stacks but define no function from stacks to trees. De Bruin and de Vink [15, 16] likewise relate a continuation semantics for Prolog with cut to an operational one without constructing the reverse translation; their proofs are on paper.

First-order unification has already been mechanized in Rocq, from educational treatments [39] and term-rewriting frameworks such as CoLoR [6] and CiME3 [14] to relational engines like miniKanren [37]. With the sole aim of keeping our mechanization axiom-free, we give a self-contained implementation of the Martelli–Montanari algorithm, extend it with matching and prove its usual properties.

8 Conclusion

This paper presents two mechanized operational semantics for Prolog with cut, together with a proof of their equivalence in Rocq. The first semantics, due to Debray and Mishra, is based on a stack of alternatives: it is close to actual Prolog implementations, including the first-order fragment of Elpi, but it leaves the scope of cut implicit in the shape of the stack, which makes it inconvenient for reasoning about cut. The second semantics is novel and based on And–Or trees, in which the scope of cut is given by the tree structure; it is well suited to stating introduction and elimination lemmas for disjunction in its presence. We show how to translate trees into stacks and use this translation to prove the two semantics equivalent. Building on the tree-based semantics, we mechanize a soundness proof for the first-order fragment of the determinacy checker of [1]: a call to a predicate identified as deterministic returns at most one solution.

The mechanization comprises approximately 2,500 lines of specifications and 5,000 lines of SSReflect proofs, built on the Mathematical Components library [29] and its finmap [31] extension. Roughly 30% of the proof code is devoted to the tree semantics, in particular the proofs in section 4.4; 15% to the stack semantics; 35% to the equivalence proof; and the remaining 20% to the determinacy checker. To keep the mechanization axiom-free, we mechanized the Marelli-Montanari unification algorithm. It amounts to about 1,700 lines of SSReflect code, including 200 lines for a termination proof and 800 lines for correctness and completeness. The mechanization took us approximately nine months.

For the static analysis of determinacy, we use a semantics in which input arguments are *matched*, rather than unified, against rule heads. This choice agrees with the Elpi interpreter, which is in turn inspired by B-Prolog's "matching clauses" [4, 46]. Our results can nonetheless be applied to a semantics based entirely on unification, by adding two assumptions to theorem 6.17: the initial query must have ground inputs, and the program must be well-moded. Under these assumptions unification (of input arguments) and matching coincide.

Use of artificial intelligence

The research and the mechanization were carried out without resorting to AI tools. The English prose was improved with the help of free LLMs.

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